10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 20 — Water

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 20.1

1. How purity of water can be checked by determining its freezing point?

A sample of water is cooled until it freezes, and then the temperature at which the ice begins to melt is noted. Pure water melts exactly at 0 °C under standard conditions. If the melting point of the sample is below 0 °C, then dissolved impurities are present in it, because impurities lower the melting point of water. In the same way, pure water boils at 100 °C, and impurities raise its boiling point, so the boiling point can also be used as a test of purity.

Quick Check 20.2

1. Why tap water is a conductor of electricity whereas distilled water is not?

Tap water contains dissolved minerals such as the salts of calcium and magnesium, and also small amounts of chlorine and fluoride which are added during treatment. These salts dissociate in water and give free ions, and the movement of these ions carries the electric current, so tap water conducts electricity. Distilled water has been purified by distillation, which removes all the dissolved salts and minerals, so it contains only water molecules and no free ions. Therefore distilled water is a bad conductor of electricity.

Quick Check 20.3

1. Which pollutants are harmful for aquatic life?

Sewage containing harmful microbes, toxic heavy metal compounds such as those of lead, mercury and cadmium, microplastics, and excess nitrates and phosphates washed from fertilizers and detergents are all harmful for aquatic life. The excess nitrates and phosphates cause the algae to grow excessively, and when this algae decays it uses up the dissolved oxygen of the water, so the fish and other aquatic organisms die of suffocation. Heavy metals and plastics also poison or choke the aquatic organisms.

2. How to identify if water is polluted?

Polluted water can often be recognised by its bad colour, unpleasant smell or taste, floating oil, foam or waste material, dead fish and excessive growth of algae. In the laboratory, chemical tests for harmful microbes, heavy metals, nitrates and pH confirm the pollution. Pure water is colourless, odourless, tasteless and neutral, with a pH of 7.

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Quick Check 20.4

1. How can we conserve water at home?

Water can be conserved at home by repairing leaking taps and pipes immediately, by closing the tap while brushing the teeth or washing dishes, and by taking short showers instead of baths. A bucket should be used instead of a running pipe for washing cars and floors, and water used for washing vegetables and clothes can be reused for watering plants. Rain water can also be collected and stored for later use.

Quick Check 20.5

1. Which is the most concentrated solid nitrogen fertilizer?

Urea is the most concentrated solid nitrogen fertilizer. It contains about 46 % nitrogen, which is more than any other solid nitrogen fertilizer. Because of its high nitrogen content, a small quantity of urea supplies a large amount of nitrogen to the crops, and that is why it is the most widely used nitrogen fertilizer in Pakistan.

Exercise (Solved)

A. Multiple Choice Questions

1. The pH of distilled water is:

(a) 6.5 (b) 8.5
(c) 7 (d) 7.5

Answer: (c) — Distilled water contains no dissolved ions, so it is exactly neutral and its pH is 7.

2. The unpleasant odours in water can be removed by:

(a) adding chlorine (b) treating it with activated charcoal
(c) sedimentation (d) filtration

Answer: (b) — Activated charcoal (carbon) adsorbs the organic compounds which are responsible for taste and odour.

3. The source of harmful microbes in natural water:

(a) human and animal wastes (b) fertilizers used for crops
(c) industrial waste-water (d) microbes in rocks

Answer: (a) — Bacteria, viruses and parasites enter water mainly through human and animal wastes, sewage and agricultural runoff.

4. Which gases are dissolved in natural water from the atmosphere through diffusion?

(a) O_2 and CO_2 (b) N_2 and CO_2
(c) O_2 and N_2 (d) O_2, N_2 and CO_2

Answer: (d) — Oxygen, nitrogen and carbon dioxide, along with other gases of the air, dissolve in natural water by diffusion.

5. The fertilizer harmful to paddy rice is:

(a) Super phosphate (b) Ammonium nitrate

(c) Urea (d) Potassium sulphate

Answer: (b) — In flooded paddy fields, the microbes decompose ammonium nitrate and convert it into nitrogen gas, which escapes and is wasted.

6. Fertilizers which increase the resistance of plants towards diseases and make plants strong by healthy root development are:

(a) Nitrogenous fertilizers (b) Phosphate fertilizers

(c) Potassium fertilizers (d) Magnesium fertilizers

Answer: (c) — Potassium forms starch, sugar and fibre, develops strong roots and increases the resistance of the plants to diseases.

B. Short Answer Questions

20.1 Name two methods to check the purity of water.

(i) Boiling point test: pure water boils at 100 °C under standard pressure, and the presence of impurities raises the boiling point. (ii) Melting point test: pure ice melts at 0 °C, and impurities lower the melting point. Another quick test is to add a drop of water to white anhydrous copper(II) sulphate, which turns blue if water is present.

20.2 What is the main difference between tap water and distilled water?

The main difference is that tap water contains dissolved minerals and salts, while distilled water contains only water molecules. Tap water is treated natural water obtained from rivers, lakes or underground sources; it contains salts of calcium and magnesium, chlorine and fluoride, and it conducts electricity because of its free ions. Distilled water is obtained by distillation, so it is free from minerals, gases and microorganisms, it has a pH of 7 and it is a bad conductor of electricity. Tap water is used for drinking and daily use, while distilled water is used in laboratories, medicines and for sterilising medical equipment.

20.3 How water is distilled in the laboratory?

Impure water is taken in a distillation flask and heated until it boils and changes into steam. The steam passes through a condenser, where cold water circulating around the inner tube cools it back into liquid water. The pure water, called distilled water, drips into the receiver, while the dissolved salts, minerals and germs, which cannot change into vapour, are left behind in the flask.

20.4 Name useful and harmful substances present in water.

Useful substances: dissolved oxygen, which fish and other aquatic organisms use for breathing, and small amounts of mineral salts which are needed by living organisms. **Harmful substances:** harmful microbes from sewage which cause diseases, toxic heavy metal compounds of lead, mercury and cadmium, microplastics, and excess nitrates and phosphates from fertilizers and detergents which cause eutrophication.

20.5 How would you remove harmful bacteria and other microbes present in water?

Harmful bacteria and other microbes are removed from water by chlorination, in which a small measured amount of liquid chlorine is added to the water; chlorine kills the germs and makes the water safe for drinking. At home, water can also be made safe by boiling it, because the high temperature kills the disease-causing microbes.

20.6 What is the difference between sedimentation and filtration?

Both processes remove suspended impurities from water, but in different ways. In sedimentation the water is allowed to stand still in large tanks for a few hours, and the heavy suspended particles settle down at the bottom by the force of gravity. In filtration the water is allowed to trickle through layers of sand and gravel, which trap the smaller suspended particles. Sedimentation removes the bigger impurities first, and filtration removes the smaller particles which are left behind.

20.7 Name any three water-borne diseases.

Cholera, typhoid and hepatitis (viral hepatitis) are three common water-borne diseases. Other water-borne diseases include dysentery, malaria, diarrhoea and trachoma, which is a bacterial eye infection that can lead to blindness.

20.8 Why fertilizers containing nitrogen element are added into the soil?

Nitrogen is needed by plants during their early stage of growth for the development of stem and leaves. It is the main constituent of the proteins of the plant, it gives the leaves their green colour, and it increases the yield and the quality of the crop. Every harvest removes nitrogen from the soil, so nitrogenous fertilizers such as urea and ammonium salts are added to the soil to make up this deficiency.

20.8 What is trachoma disease? (The book prints this question twice as 20.8.)

Trachoma is a bacterial infection of the eye which, after repeated infections, can cause blindness. It is particularly common among babies and children in developing countries and spreads through infected eye and nose secretions, dirty hands, shared cloths and flies. Clean water, regular washing of the face, and proper sanitation help to prevent this disease.

C. Constructed Response Questions

20.1 Describe the negative effects of using plastics in everyday life.

Plastic waste is not decomposed by microorganisms, so it stays in the environment for hundreds of years and pollutes the soil, rivers and seas. It breaks down into very small particles called microplastics, which have been found in tap water, bottled water and even in human lungs and blood, where they can disturb the immune and reproductive systems. Plastic bags and pieces choke fish and other aquatic animals, and plastic debris also blocks drains, which causes the flooding of cities. Burning plastic releases harmful gases into the air.

20.2 Explain one method to remove dissolved impurities present in water.

Dissolved impurities can be removed by distillation. The impure water is heated in a flask until it boils and changes into steam. The steam passes through a condenser, where it is cooled and converted back into liquid water, which is collected in the receiver. The dissolved salts, minerals and other impurities cannot change into steam, so they are left behind in the boiling flask. In this way pure distilled water is obtained.

20.3 Why urea is regarded as a good nitrogenous fertilizer?

Urea contains about 46 % nitrogen, which is the highest percentage of nitrogen in any solid nitrogen fertilizer. Because of this high nitrogen content, a small amount of urea supplies a large quantity of nitrogen to the crops, so it reduces the cost of fertilizer, storage and transport. It is readily soluble in water, it is stable in the soil, it does not change the pH of the soil, and it is suitable for most crops. These qualities make urea the most widely used nitrogenous fertilizer in Pakistan.

20.4 Why Pakistan is facing a severe shortage of water?

Pakistan lies in a semi-arid region and receives a low annual rainfall, which is mostly limited to the monsoon season. The population is growing rapidly, so the demand for water increases every year. Flood irrigation in agriculture wastes a very large quantity of water, the underground aquifers are being depleted, and climate change is disturbing the rainfall pattern and melting the glaciers. Poor water storage and weak management make the situation worse, so the country faces a serious shortage of water.

D. Descriptive Questions

20.1 How domestic water supply can be made fit for drinking?

(1) Sedimentation: the water obtained from natural sources is allowed to stand in large sedimentation tanks for a few hours, so that the heavy suspended impurities such as soil, sand and pieces of plants settle down at the bottom and form a layer of sediment. The cleaner water at the top is then taken for the next stage.

(2) Filtration: the water is allowed to trickle through beds of sand and gravel, which trap the smaller suspended particles that were not removed by sedimentation.

(3) Chlorination: a small amount of liquid chlorine is added to the water to kill the harmful bacteria and other microorganisms, so that the water becomes safe for drinking.

(4) Removal of taste and odour: the water is passed through powdered activated carbon (charcoal), which adsorbs the organic compounds responsible for bad taste and smell. Odours can also be removed by aeration, that is, by exposing the water to air.

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20.2 Which measures should be taken to avoid water-borne disease?

Only clean water should be used for drinking, and it should be filtered, boiled or treated with chlorine before use. Drinking water should be stored in clean, covered containers. Personal hygiene must be observed by keeping the body, and especially the hands, clean, and by washing hands before eating. Proper toilets, drainage and sanitation facilities should be provided and used. Food should be kept covered and away from polluted water, and patients should be treated quickly to stop the spread of infection.

20.3 What are fertilizers? Mention the essential qualities of a good fertilizer.

Fertilizers are the substances which are added to the soil to make up the deficiency of the essential elements, especially nitrogen, phosphorus and potassium (NPK), which are needed for the proper growth of plants. They enhance the natural fertility of the soil and return the elements which the previous crops have taken up from it.

Essential qualities of a good fertilizer: (i) the nutrient elements present in it must be readily available to the plants; (ii) it must be fairly soluble in water so that it mixes thoroughly with the soil; (iii) it should not be injurious to the plants; (iv) it should be inexpensive; (v) it must be stable in the soil as well as in storage, and should not absorb moisture or become hard; (vi) it should not change the pH of the soil; and (vii) with rain or irrigation water it should change into a form which the plant can easily absorb.

20.4 What are the main functions of nitrogenous, phosphatic and potassium fertilizers?

Each type of fertilizer supplies a different nutrient and performs a different function in the plant, as shown in the following table.

Type of fertilizer	Main functions
Nitrogenous fertilizers (urea, ammonium sulphate, ammonium nitrate, ammonium phosphate)	Supply nitrogen; needed in the early growth stage for the development of stem and leaves; form plant proteins; give green colour to the leaves; increase yield and quality of the crop.
Phosphatic fertilizers (super phosphate, triple phosphate/DAP)	Supply phosphorus; stimulate early growth; accelerate the formation of seeds and fruits in the later stages; increase the resistance of the plant to diseases.
Potassium fertilizers (potassium chloride, potassium sulphate, potassium nitrate)	Supply potassium; needed for the formation of starch, sugar and fibrous material; develop healthy roots and make the plant strong; help in the ripening of seeds, fruits and cereals; increase resistance to diseases.

Urea is the most concentrated solid nitrogenous fertilizer, containing about 46 % nitrogen, while ammonium nitrate is not suitable for paddy rice because the bacteria

of flooded fields decompose it into nitrogen gas.

20.5 How can you avoid the negative effects of pollutants present in water?

Sewage and industrial wastewater should be treated before it is discharged into rivers and other water bodies. Plastics should be reused or recycled to reduce their production and waste. Chemical cleaners, fats, oils and grease should never be thrown down the drain. The use of pesticides and fertilizers in agriculture should be minimised. The general public should be made aware of the harmful effects of water pollution through campaigns, and strict laws should be implemented and enforced against all types of water pollution.

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